Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

BUILDING MATERIALS AND CONSTRUCTION TECHNOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Differentiate between Load bearing structures and Framed structures. **06**

b. Write short notes on the following construction materials **08**

i. Aggregates
ii. Cement
iii. Mortar
iv. Concrete

2. a. Describe Single Flemish Bond and Double Flemish Bond with sketches. **06**

b. Explain the various types of burnt bricks and their uses. **08**

3. a. Discuss the functional requirements of the roofs in buildings. **06**

b. Discuss the factors affecting the selection of the material for flooring of a building. **08**

OR

4. a. Compare King Post truss and Queen Post roof truss with the help of neat sketches. **06**

b. Describe the various defects in painting mentioning the causes of these defects. **08**

5. a. Mention the various types of doors. Describe any three types. **07**

b. Discuss the requirements of good stairs. **07**

OR

6. a. Describe the following components of a door with sketches **06**

i. Frame
ii. Style
iii. Panels

b. With the help of sketches describe any four types of stairs classified based on the shape. **08**

7. a. What do you understand by underpinning? When do you require it? Explain the pit method of under-pinning. **08**

b. Write short notes on **06**

i. Double scaffolding

ii. Cantilever scaffolding

OR

8. a. What do you understand by shoring? Describe in brief the various types of shores. **08**

b. Discuss the requirements of good formwork. **06**